Week 9: Sales and Customer Analytics System

2. Generate Total Sales Reports Tasks: Write SQL queries to calculate:

1. Total number of orders placed.

```
SQL> SELECT COUNT(*) AS Number_Of_Orders_Placed FROM Orders;

NUMBER_OF_ORDERS_PLACED
-----------------------
                      6
```

2. Total revenue generated from sales

```
SQL> SELECT SUM(TotalAmount) AS Total_Revenue FROM Orders;

TOTAL_REVENUE
-------------
         4000
```

3. Average order value.

```
SQL> SELECT ROUND(AVG(TotalAmount),2) AS Average_Order_Value
  2  FROM Orders;

AVERAGE_ORDER_VALUE
-------------------
                620
```

4. Highest and lowest order amount.

```
SQL> SELECT MAX(TotalAmount) AS Highest_Order,
  2         MIN(TotalAmount) AS Lowest_Order
  3  FROM Orders;

HIGHEST_ORDER LOWEST_ORDER
------------- ------------
          900          400
```

5. Total sales for a specific period.

```
SQL> SELECT SUM(TotalAmount) AS Total_Revenue
  2  FROM Orders
  3  WHERE OrderDate BETWEEN DATE '2026-06-01'
  4  AND DATE '2026-08-31';

TOTAL_REVENUE
-------------
         1000
```

3. Customer Purchase Analysis

1. Total number of orders placed by each customer.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS TOTAL_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_ID
--------------------
CUSTOMER_NAME                                              TOTAL_ORDERS
---------------------------------------------------------- ------------
CUST000001
Arjun Kumar                                                           3

CUST000003
Rahul Menon                                                           1

CUST000004
Ananya Iyer                                                           1


CUSTOMER_ID
--------------------
CUSTOMER_NAME                                              TOTAL_ORDERS
---------------------------------------------------------- ------------
CUST000002
Priya Sharma                                                          1

CUST000005
Vikram Reddy                                                          0
```

2. Total amount spent by each customer.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      NVL(SUM(O.TOTALAMOUNT), 0) AS TOTAL_AMOUNT_SPENT
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_AMOUNT_SPENT DESC;

CUSTOMER_ID
--------------------
CUSTOMER_NAME                                                TOTAL_AMOUNT_SPENT
------------------------------------------------------------ ------------------
CUST000001
Arjun Kumar                                                                2500

CUST000002
Priya Sharma                                                                700

CUST000003
Rahul Menon                                                                 500


CUSTOMER_ID
--------------------
CUSTOMER_NAME                                                TOTAL_AMOUNT_SPENT
------------------------------------------------------------ ------------------
CUST000004
Ananya Iyer                                                                 400

CUST000005
Vikram Reddy                                                                  0
```

3. Average spending per customer.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      ROUND(NVL(AVG(O.TOTALAMOUNT), 0), 2) AS AVERAGE_SPENDING
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY AVERAGE_SPENDING DESC;

CUSTOMER_ID
--------------------
CUSTOMER_NAME                                              AVERAGE_SPENDING
---------------------------------------------------------- ----------------
CUST000001
Arjun Kumar                                                          833.33

CUST000002
Priya Sharma                                                           700

CUST000003
Rahul Menon                                                            500


CUSTOMER_ID
--------------------
CUSTOMER_NAME                                              AVERAGE_SPENDING
---------------------------------------------------------- ----------------
CUST000004
Ananya Iyer                                                            400

CUST000005
Vikram Reddy                                                             0
```

4. Customers with maximum purchase amount.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTALAMOUNT) AS TOTAL_PURCHASE_AMOUNT
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING SUM(O.TOTALAMOUNT) = (
 13      SELECT MAX(TOTAL_SPENDING)
 14      FROM (
 15          SELECT SUM(TOTALAMOUNT) AS TOTAL_SPENDING
 16          FROM ORDERS
 17          GROUP BY CUSTOMERID
 18      )
 19  );

CUSTOMER_ID
--------------------
CUSTOMER_NAME
--------------------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
CUST000001
Arjun Kumar
                2500
```

5. Customers with fewer purchases.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDERID) <= 1
 13  ORDER BY NUMBER_OF_ORDERS;

CUSTOMER_ID
--------------------
CUSTOMER_NAME                                                   NUMBER_OF_ORDERS
--------------------------------------------------------------- ----------------
CUST000005
Vikram Reddy                                                                   0

CUST000004
Ananya Iyer                                                                    1

CUST000002
Priya Sharma                                                                   1


CUSTOMER_ID
--------------------
CUSTOMER_NAME                                                   NUMBER_OF_ORDERS
--------------------------------------------------------------- ----------------
CUST000003
Rahul Menon                                                                    1
```

4. Find Top Customers Based on Purchase Amount

1. Top 5 customers based on total spending.

```
SQL> SELECT *
  2  FROM (
  3      SELECT
  4          C.CUSTOMER_ID,
  5          C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  6          SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  7      FROM CUSTOMER C
  8      JOIN ORDERS O
  9          ON C.CUSTOMER_ID = O.CUSTOMERID
 10      GROUP BY
 11          C.CUSTOMER_ID,
 12          C.FIRST_NAME,
 13          C.LAST_NAME
 14      ORDER BY TOTAL_SPENDING DESC
 15  )
 16  WHERE ROWNUM <= 5;

CUSTOMER_ID
--------------------
CUSTOMER_NAME                                              TOTAL_SPENDING
---------------------------------------------------------- --------------
CUST000001
Arjun Kumar                                                          2500

CUST000002
Priya Sharma                                                          700

CUST000003
Rahul Menon                                                           500


CUSTOMER_ID
--------------------
CUSTOMER_NAME                                              TOTAL_SPENDING
---------------------------------------------------------- --------------
CUST000004
Ananya Iyer                                                           400
```

2. Customers with maximum number of orders.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDERID) = (
 13      SELECT MAX(ORDER_COUNT)
 14      FROM (
 15          SELECT COUNT(*) AS ORDER_COUNT
 16          FROM ORDERS
 17          GROUP BY CUSTOMERID
 18      )
 19  )
 20  ORDER BY CUSTOMER_NAME;

CUSTOMER_ID
--------------------
CUSTOMER_NAME                                              NUMBER_OF_ORDERS
---------------------------------------------------------- ----------------
CUST000001
Arjun Kumar                                                               3
```

3. Frequent customers.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDERID) > 1
 13  ORDER BY NUMBER_OF_ORDERS DESC;

CUSTOMER_ID
--------------------
CUSTOMER_NAME                                              NUMBER_OF_ORDERS
---------------------------------------------------------- ----------------
CUST000001
Arjun Kumar                                                               3
```

4. High-value customers.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING SUM(O.TOTALAMOUNT) > 1000
 13  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_ID
--------------------
CUSTOMER_NAME                                              TOTAL_SPENDING
---------------------------------------------------------- --------------
CUST000001
Arjun Kumar                                                          2500
```

5. Identify Best-Selling Products

1. Products with maximum sales quantity

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_SALES_QUANTITY
  5  FROM PRODUCT P
  6  JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  HAVING SUM(OD.QUANTITY) = (
 12      SELECT MAX(TOTAL_QUANTITY)
 13      FROM (
 14          SELECT SUM(QUANTITY) AS TOTAL_QUANTITY
 15          FROM ORDERDETAIL
 16          GROUP BY PRODUCTID
 17      )
 18  )
 19  ORDER BY P.PRODUCT_NAME;

PRODUCT_ID           PRODUCT_NAME
-------------------- --------------------------------------------------
TOTAL_SALES_QUANTITY
--------------------
P104                 Lipstick
                   3

P102                 Moisturizer
                   3
```

2. Products generating highest revenue.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  5      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  6  FROM PRODUCT P
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      P.PRODUCT_ID,
 11      P.PRODUCT_NAME
 12  ORDER BY TOTAL_REVENUE DESC;

PRODUCT_ID           PRODUCT_NAME
-------------------- ------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
P102                 Moisturizer
            3          1050

P105                 Sunscreen
            2          1000

P104                 Lipstick
            3           600


PRODUCT_ID           PRODUCT_NAME
-------------------- ------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
P103                 Shampoo
            1           300

P101                 Face Wash
            1           250
```

3. Least-selling products.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD
  5  FROM PRODUCT P
  6  LEFT JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY QUANTITY_SOLD ASC;

PRODUCT_ID           PRODUCT_NAME
-------------------- ----------------------------------------
QUANTITY_SOLD
-------------
P101                 Face Wash
             1

P103                 Shampoo
             1

P105                 Sunscreen
             2


PRODUCT_ID           PRODUCT_NAME
-------------------- ----------------------------------------
QUANTITY_SOLD
-------------
P102                 Moisturizer
             3

P104                 Lipstick
             3
```

4. Products requiring promotion.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      P.PRICE,
  5      P.STOCK_QUANTITY,
  6      NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD
  7  FROM PRODUCT P
  8  LEFT JOIN ORDERDETAIL OD
  9      ON P.PRODUCT_ID = OD.PRODUCTID
 10  GROUP BY
 11      P.PRODUCT_ID,
 12      P.PRODUCT_NAME,
 13      P.PRICE,
 14      P.STOCK_QUANTITY
 15  HAVING NVL(SUM(OD.QUANTITY), 0) <= 1
 16  ORDER BY QUANTITY_SOLD, P.PRODUCT_NAME;

PRODUCT_ID           PRODUCT_NAME
-------------------- ------------------------------------
     PRICE STOCK_QUANTITY QUANTITY_SOLD
---------- -------------- -------------
P101                 Face Wash
       250             50             1

P103                 Shampoo
       300             30             1
```

6. Perform Category-wise Sales Analysis

1. Total sales generated by each category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_SALES
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY TOTAL_SALES DESC;

CATEGORY_NAME                                      TOTAL_SALES
-------------------------------------------------- -----------
Skincare                                              2300
Makeup                                                 600
Haircare                                               300
```

2. Number of products sold in each category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY) AS PRODUCTS_SOLD
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY PRODUCTS_SOLD DESC;

CATEGORY_NAME                                      PRODUCTS_SOLD
-------------------------------------------------- -------------
Skincare                                                       6
Makeup                                                         3
Haircare                                                       1
```

3. Highest revenue-generating category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  HAVING SUM(OD.QUANTITY * OD.PRICE) = (
 13      SELECT MAX(CATEGORY_REVENUE)
 14      FROM (
 15          SELECT SUM(OD2.QUANTITY * OD2.PRICE) AS CATEGORY_REVENUE
 16          FROM CATEGORY C2
 17          JOIN PRODUCT P2
 18              ON C2.CATEGORY_ID = P2.CATEGORY_ID
 19          JOIN ORDERDETAIL OD2
 20              ON P2.PRODUCT_ID = OD2.PRODUCTID
 21          GROUP BY C2.CATEGORY_ID
 22      )
 23  );

CATEGORY_NAME                                      TOTAL_REVENUE
-------------------------------------------------- -------------
Skincare                                                    2300
```

4. Average sales per category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      ROUND(AVG(OD.QUANTITY * OD.PRICE), 2) AS AVERAGE_SALES
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY AVERAGE_SALES DESC;

CATEGORY_NAME                                      AVERAGE_SALES
-------------------------------------------------- -------------
Skincare                                                     460
Haircare                                                     300
Makeup                                                       300
```

7. Generate Business Analytics Reports

Report 1: Sales Performance Report

• Total sales

• Number of orders

• Average order value

Highest order value

```
SQL> SELECT
  2      SUM(TOTALAMOUNT) AS TOTAL_SALES,
  3      COUNT(ORDERID) AS NUMBER_OF_ORDERS,
  4      ROUND(AVG(TOTALAMOUNT), 2) AS AVERAGE_ORDER_VALUE,
  5      MAX(TOTALAMOUNT) AS HIGHEST_ORDER_VALUE
  6  FROM ORDERS;

TOTAL_SALES NUMBER_OF_ORDERS AVERAGE_ORDER_VALUE HIGHEST_ORDER_VALUE
----------- ---------------- ------------------- -------------------
       4100                6              683.33                1000
```

Report 2: Customer Analytics Report

• Customer name

• Number of orders

• Total spending

• Purchase frequency

```
SQL> SELECT
  2      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS,
  4      NVL(SUM(O.TOTALAMOUNT), 0) AS TOTAL_SPENDING,
  5      COUNT(O.ORDERID) AS PURCHASE_FREQUENCY
  6  FROM CUSTOMER C
  7  LEFT JOIN ORDERS O
  8      ON C.CUSTOMER_ID = O.CUSTOMERID
  9  GROUP BY
 10      C.CUSTOMER_ID,
 11      C.FIRST_NAME,
 12      C.LAST_NAME
 13  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_NAME                                                  NUMBER_OF_ORDERS
-------------------------------------------------------------- ----------------
TOTAL_SPENDING PURCHASE_FREQUENCY
-------------- ------------------
Arjun Kumar                                                                   3
          2500                  3

Priya Sharma                                                                  1
           700                  1

Rahul Menon                                                                   1
           500                  1


CUSTOMER_NAME                                                  NUMBER_OF_ORDERS
-------------------------------------------------------------- ----------------
TOTAL_SPENDING PURCHASE_FREQUENCY
-------------- ------------------
Ananya Iyer                                                                   1
           400                  1

Vikram Reddy                                                                  0
             0                  0
```

Report 3: Product Performance Report

- Product name
- Quantity sold
- Revenue generated

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS REVENUE_GENERATED
  5  FROM PRODUCT P
  6  JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY REVENUE_GENERATED DESC;

PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Moisturizer                                                    3
             1050

Sunscreen                                                      2
             1000

Lipstick                                                       3
              600


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Shampoo                                                        1
              300

Face Wash                                                      1
              250
```

Report 4: Category Analysis Report

- Category name
- Total products sold
- Total revenue

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY) AS TOTAL_PRODUCTS_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  5  FROM CATEGORY C
  6  JOIN PRODUCT P
  7      ON C.CATEGORY_ID = P.CATEGORY_ID
  8  JOIN ORDERDETAIL OD
  9      ON P.PRODUCT_ID = OD.PRODUCTID
 10  GROUP BY
 11      C.CATEGORY_ID,
 12      C.CATEGORY_NAME
 13  ORDER BY TOTAL_REVENUE DESC;

CATEGORY_NAME                                      TOTAL_PRODUCTS_SOLD
-------------------------------------------------- -------------------
TOTAL_REVENUE
-------------
Skincare                                                             6
         2300

Makeup                                                               3
          600

Haircare                                                             1
          300
```